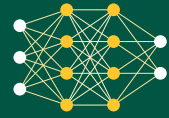


Neural Networks

Fall 2026 • 3 credits • North Dakota State University



Derive deep learning from first principles, implement every piece in PyTorch, and finish the semester with a working model on a problem you chose.

What You Will Learn

Foundations

Shallow and deep networks, what depth actually buys you, backpropagation worked through by hand, and initialization that keeps gradients alive.

Training that works

Loss functions from maximum likelihood, stochastic gradient descent and Adam, regularization, and how to measure generalization honestly.

Modern architectures

Convolutional networks, transformers and the attention mechanism, large language models, and graph neural networks for relational data.

Generative models

Variational autoencoders, generative adversarial networks, and the diffusion models behind today's image generators.

How the Course Runs

- Every topic pairs the mathematics with a runnable PyTorch implementation — no black boxes.
- Roughly weekly assignments, distributed and submitted as Jupyter notebooks.
- A semester project on a real problem. Bring your own research data and leave with a result you can publish.
- Read and present a recent paper, then argue about where the field goes next.

Who Should Enroll

Graduate students in **engineering** and **computer science**, and anyone whose research runs on data. The only prerequisite is graduate standing — no previous machine learning course is required. You should be comfortable with linear algebra, calculus, and basic Python; the course builds the rest.

MEETS

Tuesday & Thursday
12:30–1:45 p.m.

TERM

August 25 – December 10, 2026

OPEN TO

All graduate students

PREREQUISITE

Graduate standing

SOFTWARE

Python and PyTorch.

Google Colab is fine — nothing to install.

TEXTBOOK

Prince, *Understanding Deep Learning*, MIT Press, 2024 — **free online**.

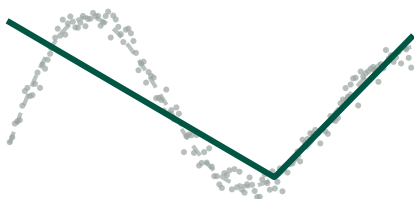


Syllabus, schedule, and every lecture note — free and open.

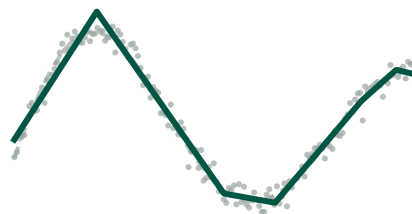
harunpirim.github.io/IMSE774-NeuralNetworks

One hidden layer, one nonlinearity, and enough units: the universal approximation theorem, made concrete in week 3.

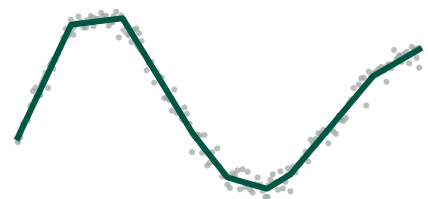
3 hidden units



10 hidden units



50 hidden units



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Questions? Email me.

Register through NDSU
Campus Connection.